



- 5 A particle P of mass m kg is projected vertically upwards from a point O with an initial speed of 20 m s^{-1} and moves under gravity. There is a resistive force of magnitude $0.025mv^2 \text{ N}$, where $v \text{ m s}^{-1}$ is the speed of P at time t s after projection. The displacement of P from O is x m at time t s after projection.

(a) Find an expression for v in terms of x , while P is moving upwards. [5]

[illegible]



(b) Find an expression for v in terms of t , while P is moving upwards.

[4]

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